

Noise and Outliers

Project: Transparent Book Recommender

1. Noise and Outliers

Teach the learning cycle

Primary target: Calculate mean and median

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Overfitting Intuition

Without opening the old lesson, explain one key idea from Overfitting Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare training and validation evidence for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf predictions from training examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one. Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Training and Testing

Without opening the old lesson, explain one key idea from Training and Testing. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended prediction claim

Reference example, not the only acceptable wording: Training fits the model; validation compares development choices; test evaluates after choices are fixed. Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

Concept to explain

Mistakes and unusual values should be investigated, not automatically erased.

Term · Meaning

Term: noise

Meaning: Variation or interference that obscures the information relevant to a task.

Term: outlier

Meaning: An observation far from a typical pattern under a stated comparison; it is not automatically an error.

Term: duplicate

Meaning: A repeated record or copy; copies are not independent new observations.

Term: investigate

Meaning: Collect and examine evidence to answer a question rather than assuming a cause.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 17 as a two-session investigation using fictional data. Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects. Keep all values while investigating whether 18 is a genuine long activity or a recording error. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

9.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can calculate mean and median and investigate an unusual value before changing it.

This added worked case illustrates the week's main idea. Retain all original and connected topics;

schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Most books: 40–300 pages; one record: 3,200 pages.

2. Check the source: extra zero, box set, or real long work?

3. Repair confirmed error; keep real rare case and document it.

4. Retest to learn whether the system handles legitimate extremes.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

Exact explanation to model

No. It is an outlier, but may be a real large book. Check catalogue, units and entry process.

Explain the added worked example

Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects. Keep all values while investigating whether 18 is a genuine long activity or a recording error.

The script prints 6.0 and 3. `sorted` returns a sorted list without changing the original. This median expression works for a nonempty odd-length list only. For four sorted values, average the second and third. Do not use this odd-length shortcut for every dataset.

Key terms

Term · Meaning

Term: Mean

Meaning: Sum of numeric values divided by their count

Term: Median

Meaning: The middle of sorted values or mean of the two middle values

Term: Unusual value

Meaning: A value far from most others in a dataset

Term: Variation

Meaning: Differences among recorded values

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Replace 18 with 8. Calculate the new mean and median.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Noise and Outliers

A fictional library records five book-processing times in minutes.

The facts we will use

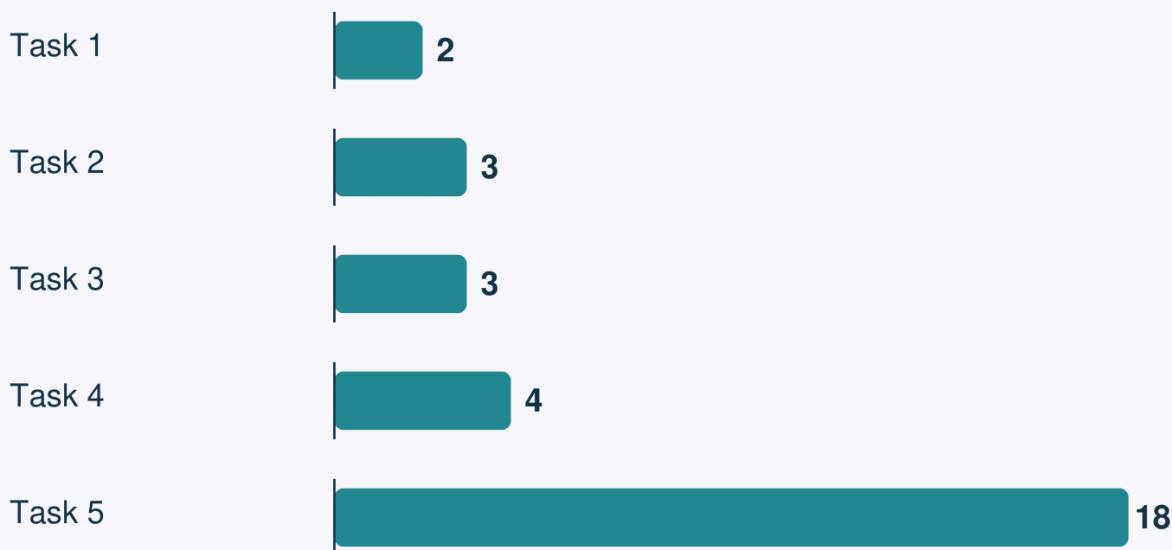
Sorted times: 2,3,3,4,18.

All are recorded measurements; no evidence yet shows 18 is a mistake.

AI ACADEMY / AI-2/17

Retain unusual recorded measurements

minutes



Bar length shows the supplied value; the marked baseline is zero.

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Mean

What happens and why: Sum 30; divide by 5: mean 6 minutes.

Step: 2. Median

What happens and why: The middle of five sorted values is 3 minutes.

Step: 3. Investigate 18

What happens and why: Check the source record and whether an unusually slow task occurred.

Step: 4. Document repair

What happens and why: Change a value only if evidence supports a correction; preserve the raw record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The mean uses every value and is pulled upward by 18. The median identifies the middle recorded time and stays at 3. Neither statistic declares the unusual observation invalid. Deleting it simply because it changes the mean would hide potentially real slow work and change the question being answered.

What this example does not establish

Five task times do not estimate all library processing times reliably or prove which summary suits every decision.

Change one thing in the pictured case

Change: An authorised log confirms 18 was a transcription error for 3.

Prediction: Corrected mean is 3 minutes and median is 3.

Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Recorded lengths are 90,95,100,105,900 pages.

Write the question to explain: Is 900 automatically wrong?

Reveal after predictions: No. It is an outlier, but may be a real large book. Check catalogue, units and entry process.

Keep for an independent attempt: Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to investigate.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Sum 20; mean 4; median remains 3. Exit: No. Zero would be a measured summary; no numeric observations are available.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: Find medians of [9, 1, 3] and [8, 2, 4, 6].

Teacher check: Sorted first list 1, 3, 9 has median 3. Sorted second 2, 4, 6, 8 has median $(4 + 6)/2 = 5$.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Mean median and unusual values. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Removing 18 merely because it raises the mean would hide evidence. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Reasoned teacher answer

Median 3 describes the middle; also report the unusual 18-minute task and the mean 6 rather than concealing spread.

Changed-case reasoning: Corrected mean is 3 minutes and median is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Audit four suspicious records using keep / correct / remove / ask labels. Require a reason and a source check for every action.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Removing 18 merely because it raises the mean would hide evidence. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: 2, 4, 4, 5, 15 has sum 30, mean 6 and median 4. Replacing 15 with 5 gives sum 20, mean 4 and median 4.

A convenient shortcut preserves meaning: Taking the middle item of an unsorted list can give the wrong median. For [18, 2, 4, 3, 3], the middle item is 4, but the sorted median is 3. Sort first. For an empty list, there is no mean or median under this lesson's rules.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

No. It is an outlier, but may be a real large book. Check catalogue, units and entry process.

Independent question

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to investigate.

Expected independent answer

900 may be a genuine long book or a recording error. Check the original record and task scope before changing it; unusual does not automatically mean wrong.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: "Every outlier is an error." Keep the first attempt and an uncoached retry separate.

Original transfer answer

Check edition, catalogue and source entry. Retain 960 if verified; correct only if evidence establishes an error, recording original value, new value and reason. Automatic deletion can remove a legitimate category and distort the intended collection.

Explanations for the additional workbook examples

W1 Compute a mean

Find the mean of 4, 4, 6, 6 and 10.

Sum 30 divided by 5 gives 6.

Add all five observations: $4 + 4 + 6 + 6 + 10 = 30$. Divide by the number of observations, five, to obtain a mean of 6. Keep the count beside the calculation. Do not divide by the three distinct values; repeated observations still count.

W2 Compute two medians

Find medians of [9, 1, 3] and [8, 2, 4, 6].

Sorted first list 1, 3, 9 has median 3. Sorted second 2, 4, 6, 8 has median $(4 + 6)/2 = 5$.

W3 Repair an explanation

A learner deletes 18 because it is above the mean. What evidence is needed before correcting or excluding it?

Check original measurement and collection rules. Being above the mean alone is not an error. Retain the raw record and document any justified correction.

W4 Compare summaries independently

For 1, 2, 2, 3 and 12, calculate mean and median. Then replace 12 with 2 and recalculate. Explain the change.

Original sum 20, mean 4, median 2. Revised sum 10, mean 2, median 2. The large value changes mean more than median in this example.

W5 Report missing evidence

A dataset is empty. Is reporting a mean of 0 justified?

No. Zero would be a measured summary; no numeric observations are available.

Assess independent transfer

Selected case: For 1, 2, 2, 3 and 12, calculate mean and median. Then replace 12 with 2 and recalculate. Explain the change.

Selected case answer

Original sum 20, mean 4, median 2. Revised sum 10, mean 2, median 2. The large value changes mean more than median in this example.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Keep legitimate unusual books so the recommender can serve unusual readers.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 18: Missing Data. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Order number cards physically. Extension: write a median function with explicit empty, odd and even cases and test each.

Independent W4 evidence: Original sum 20, mean 4, median 2. Revised sum 10, mean 2, median 2. The large value changes mean more than median in this example. Record whether the learner needed no help, a hint or explanation. Fresh retry: 2, 4, 4, 5, 15 has sum 30, mean 6 and median 4. Replacing 15 with 5 gives sum 20, mean 4 and median 4.

Next week: Graphs and honest scales. Reuse this week's explicit contracts and evidence.

Plan the delayed return

Suggested return points in this level: ai-2/18, ai-2/20, ai-2/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Investigate unusual recommender features and show evidence-backed repairs instead of removing inconvenient records.

7. Prepare, teach and assess

Teaching target: Calculate mean and median

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Find medians of [9, 1, 3] and [8, 2, 4, 6].

Reasoned practice answer: Sorted first list 1, 3, 9 has median 3. Sorted second 2, 4, 6, 8 has median $(4 + 6)/2 = 5$.

Check every learner before discussion: Use the worked case for Mean median and unusual values. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Removing 18 merely because it raises the mean would hide evidence. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: An authorised log confirms 18 was a transcription error for 3. Require a prediction and a reason before comparing. Expected result: Corrected mean is 3 minutes and median is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.